Embeddings: a Map of Meaning

Module 4 · Zero to LLM

IDs are seat numbers

Seat 3290 isn't "more dog" than seat 3291.

We need **coordinates**.

One-hot vs dense

one-hot	dense embedding
50 257 numbers, one `1`	768 learned numbers
all words equally far	similar words are neighbours

"A word is known by the company it keeps."

The table

`vocab × d_model` matrix → lookup = take a row

GPT-2: 50 257 × 768 ≈ **38.6 M** parameters

Cosine similarity = angle between arrows

⚠ `"king"` (no space) ≠ `" king"` — embed the token the model really sees

Order matters: positions

method	used by
learned table	GPT-2, BERT
fixed sinusoids	2017 Transformer
rotary (RoPE)	Llama, Mistral, Qwen…

```
x = token_emb + pos_emb
```

Next: M05 — Attention